

B. Tech Degree IV Semester (Supplementary) Examination January 2011

ME 403 APPLIED THERMO DYNAMICS (2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) State and explain Kelvin-Plank's statement and Clausius's statement for second law of Thermodynamics. Also prove that above said statements are equivalent to each other. (10)
- (b) Consider an engine to be working in outerspace on carnot cycle. The only way to transfer the heat to engine is radiation which obeys the law $Q = \sigma AT^4$ (where σ – constant, A – area of radiating surface and T the temperature of body in absolute scale). Show that for a given power output and a given temperature (T_1) the area of radiator will be minimum when, $\frac{T_2}{T_1} = \frac{3}{4}$. (10)
- OR**
- II. (a) Explain the concept of availability. Obtain availability for a steady flow process and non flow process. (10)
- (b) Exhaust gases leave an internal combustion engine at 800°C and later, after having done 1050 kJ if work per kg of gas in the engine. (C_p of gas = 1.1 kJ/kgK). The temperature of the surroundings is 30°C .
- (i) How much available energy per kg of the gas is lost by throwing away the exhaust gases?
- (ii) What is the ratio of cost available energy to engine work. (10)
- III. (a) Explain Datton's law of partial pressure and Amagat's law of partial volume for perfect gases. (10)
- (b) A perfect gas mixture consists of 1.4 kg of N_2 and 2.3 kg of CO_2 at a pressure of 2.8 bar and a temperature of 20°C . Find:
- (i) Mole fraction of each constituent
- (ii) Equivalent molecular weight of mixture.
- (iii) Equivalent gas constant of the mixture.
- (iv) Partial pressure and partial volume.
- (v) Volume and density of the mixture. (10)
- OR**
- IV. (a) State and derive Gibb's – Datton law of mixtures of ideal gases. (10)
- (b) The volumetric analysis of a gas is as follows,

CO_2 – 40%

N_2 – 40%

O_2 – 20%

The pressure of the gas is 1 bar and temperature is 15°C . Calculate the partial pressures of each constituent. (10)

- V. (a) Explain the experimental procedure for determining the calorific value of fuels using a neat sketch. (10)
- (b) The product of combustion of a hydro carbon fuel of unknown composition have the following product composition as measured on dry basis.
 $\text{CO}_2 = 8\%$, $\text{CO} = 0.9\%$, $\text{O}_2 = 8.8\%$, $\text{N}_2 = 82.3\%$

Calculate the air fuel ratio and percentage of theoretical air on a mass basis. (10)

OR

- VI. (a) Explain the concept of Adiabatic flame temperature and the factors effecting it. (5)
- (b) Propane is burnt with 200% theoretical air. Find Adiabatic flame temperature in this case. Also find the change in the adiabatic flame temperature when 200% excess air is used. (15)

- VII. (a) Compare otto cycle, diesel cycle and dual cycle on ideal conditions for same heat input and compression ratio with neat sketches of PV & TS diagrams. (10)
- (b) Explain the effect of dissociation and variable specific speeds on performance of heat engines. (10)

OR

- VIII. (a) Explain different scavenging methods adopted in 2-stroke petrol engines. (10)
- (b) A 4-cylinder engine was tested at constant speed and throttle opening. The fuel supply was 0.089 kg/minute where all cylinders were firing. The spark plugs were then successively short circuited without change and speed. The brake power measurements were as follows:

With all cylinders firing	-	15kw
Cylinder 1 cut out	-	9.04 kw
Cylinder 2 cut out	-	9.3 kw
Cylinder 3 cut out	-	9.32 kw
Cylinder 4 cut out	-	9.26 kw

Determine the mechanical efficiency and indicated thermal efficiency of the engine if calorific value of fuel used is 41, 840 kJ/kg. (10)

- IX. With neat sketches explain different combustion chambers used in SI engines. Also cite their properties. (20)

OR

- X. (a) Explain Wankel engine with a neat sketch. (10)
- (b) Explain battery ignition system and magneto ignition system with a neat sketch. (10)